

Formula Sheet for Chapter 8

$$\text{Expected Return} = \sum (\text{Probability of Outcome} \times \text{Return of Outcome})$$

Measure Risk using standard deviation

- From a Probability Distribution. $\sigma = \sqrt{\sum [P_i \times (r_i - \hat{r})^2]}$

- From Historical Data $\sigma = \sqrt{\sum \frac{(r_i - \hat{r})^2}{n}}$

In probability distribution, fatter or wider distribution → Higher Risk

Coefficient of Variation (CV)	$CV = \frac{\sigma}{\hat{r}}$
Sharpe Ratio	$SHR = \frac{\hat{r} - r_f}{\sigma}$
Expected Portfolio Return	$\hat{r}_p = \sum_{i=1}^n w_i \times \hat{r}_i$
CAPM Model Or Required rate of return	$r_i = r_f + \beta_i \times (r_m - r_f)$

Beta (β) measures a stock's sensitivity to movements in the overall market.

$\beta = 1$ → Same volatility as the market → Market risk

$\beta > 1$ → More volatile than the market → Higher risk

$\beta < 1$ → Less volatile than the market → Lower risk

$\beta < 0$ → Moves opposite to the market → Hedge characteristics

